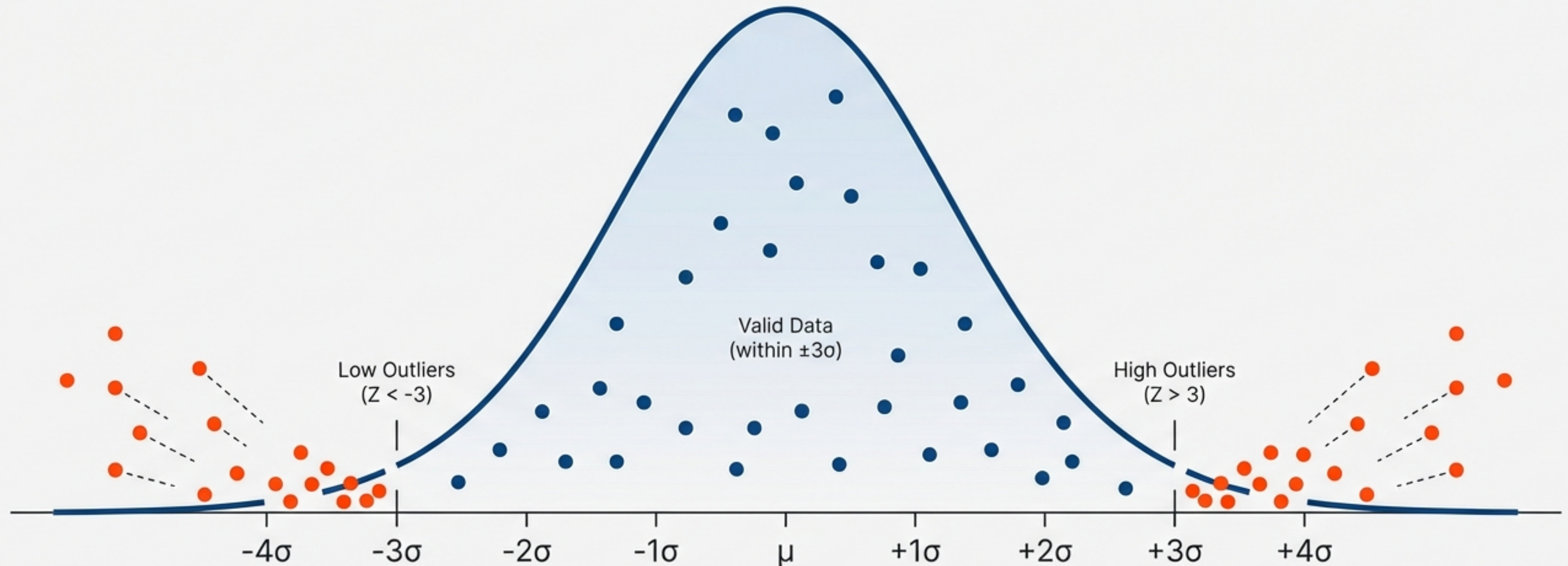


Outlier Detection & Removal: The Z-Score Method

A statistical approach to handling anomalies in normally distributed data.



The Prerequisite: Is Your Data Normal?

The Constraint:

This technique is strictly limited to data features that follow an approximate Normal Distribution.

Characteristics:

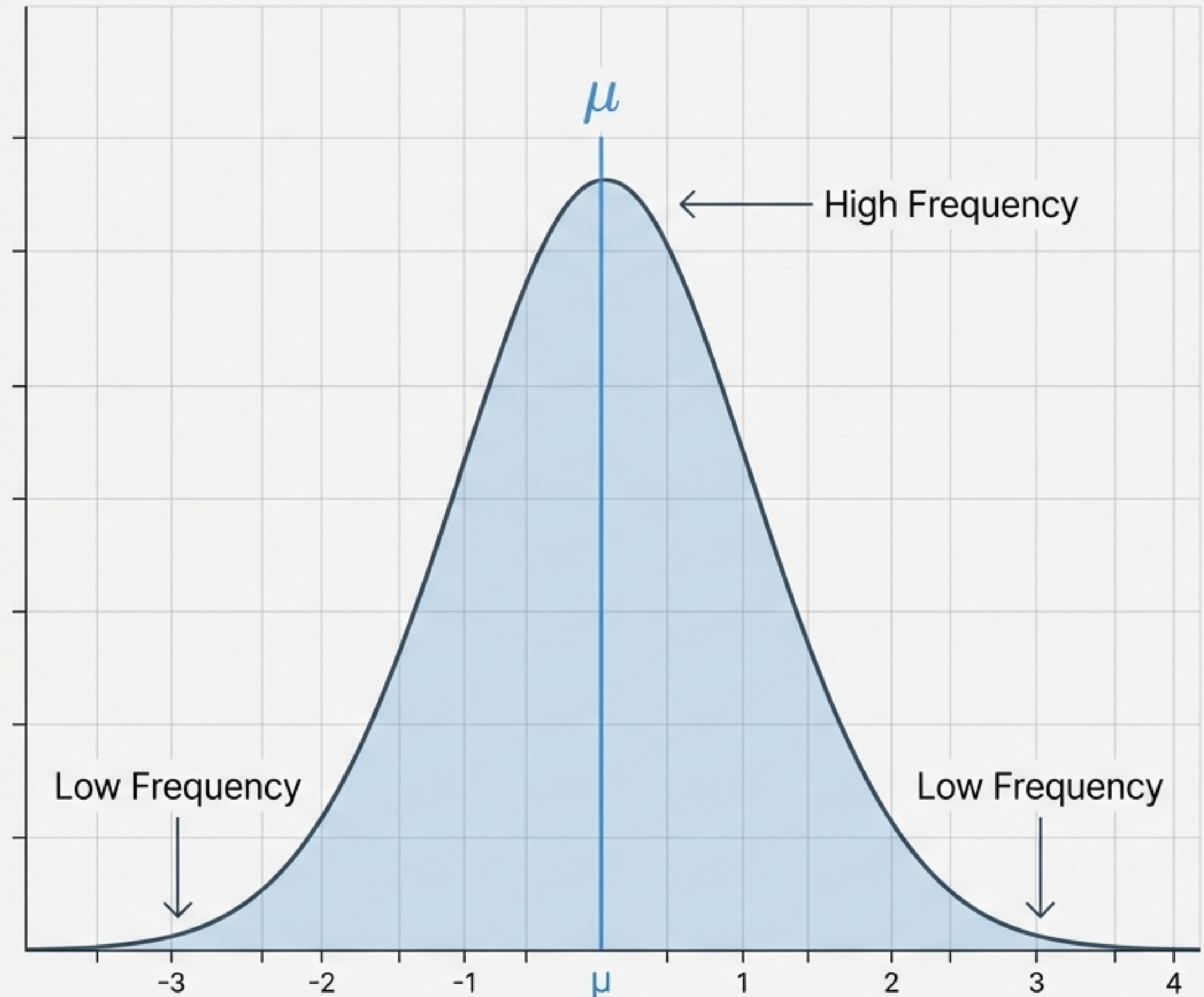
Most values cluster around the mean. Frequency decreases symmetrically as you move away from the centre.

Common Examples:

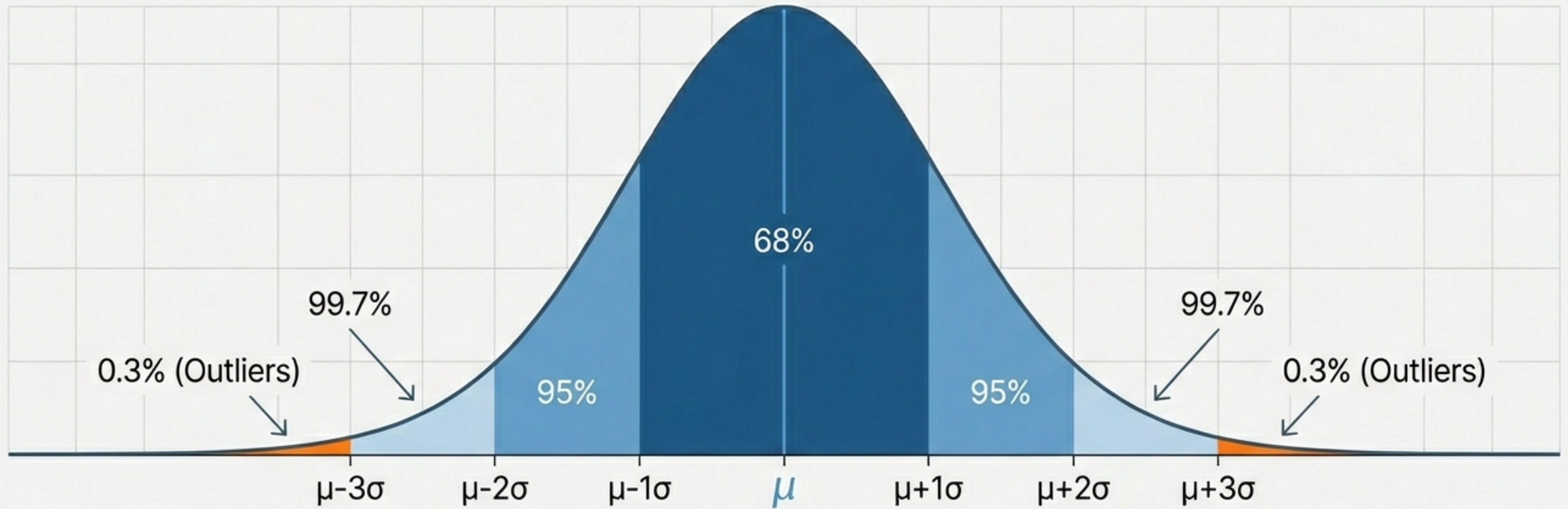
Human Height, Weight, Standardised Exam Results.

Critical Warning:

Do not use this method on skewed data.

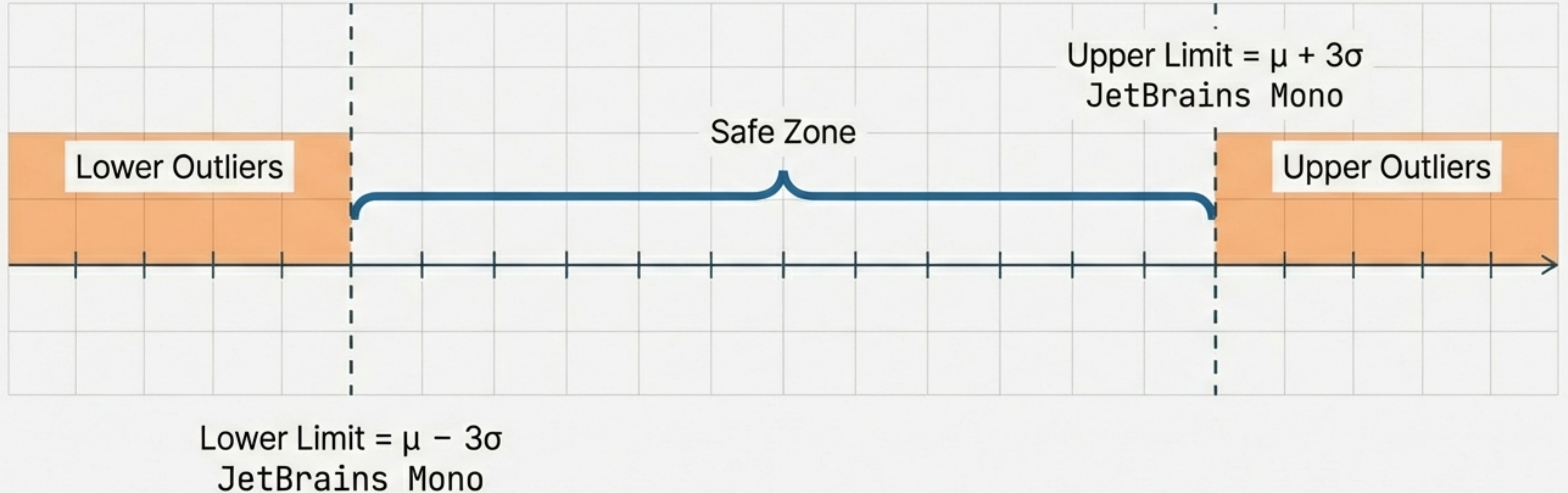


Trusting the Math: The Empirical Rule



The 68-95-99.7 Rule states that 99.7% of all data points lie within 3 Standard Deviations (σ) of the Mean (μ). Therefore, any value outside this boundary is statistically rare.

Defining the Boundaries

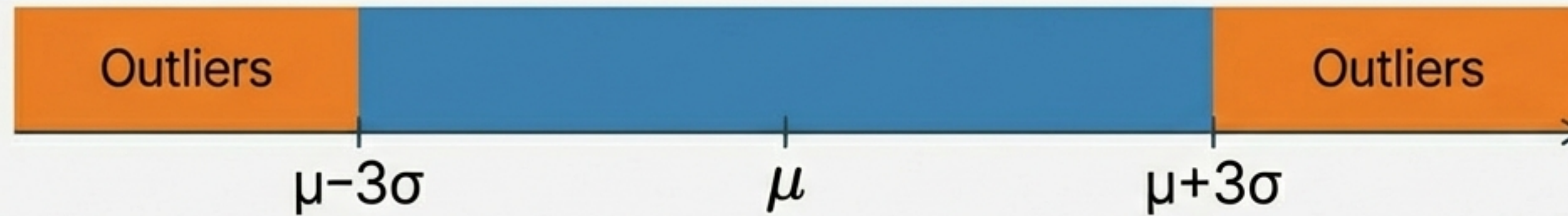


If Value > ($\mu + 3\sigma$) OR Value < ($\mu - 3\sigma$) → Outlier

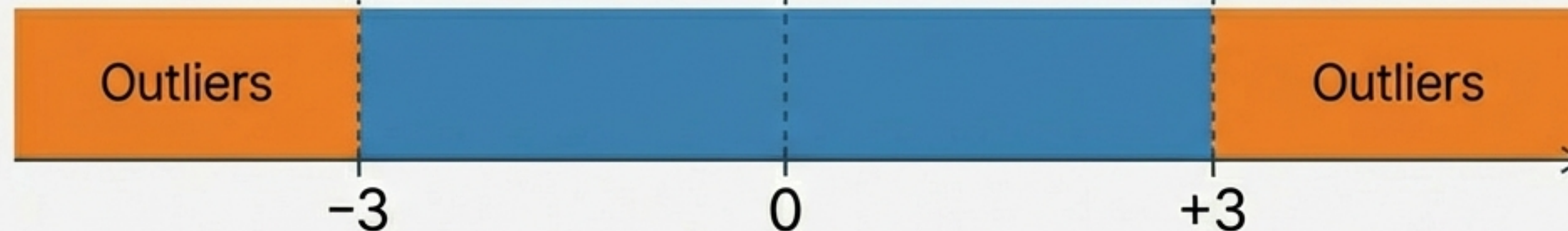
Why is it called the 'Z-Score' Method?

A Z-score standardises data, telling us how many standard deviations a point is from the mean. Calculating the Range Limits is that is mathematically identical to checking Z-scores.

Raw Data Distribution



Standardised Z-Score



$$Z = \frac{X - \mu}{\sigma}$$

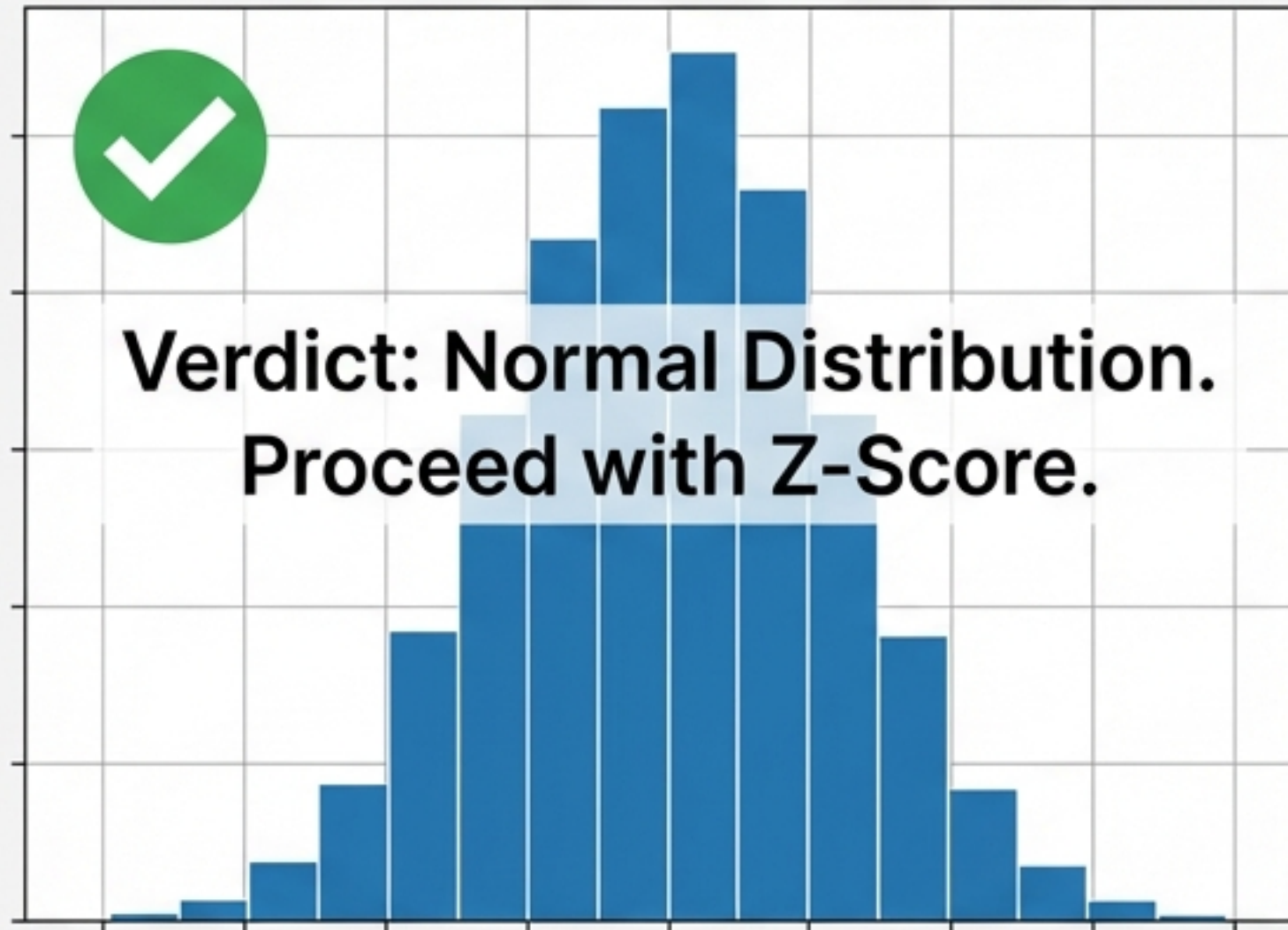
Outlier Criterion:

$$Z > 3 \text{ or } Z < -3$$

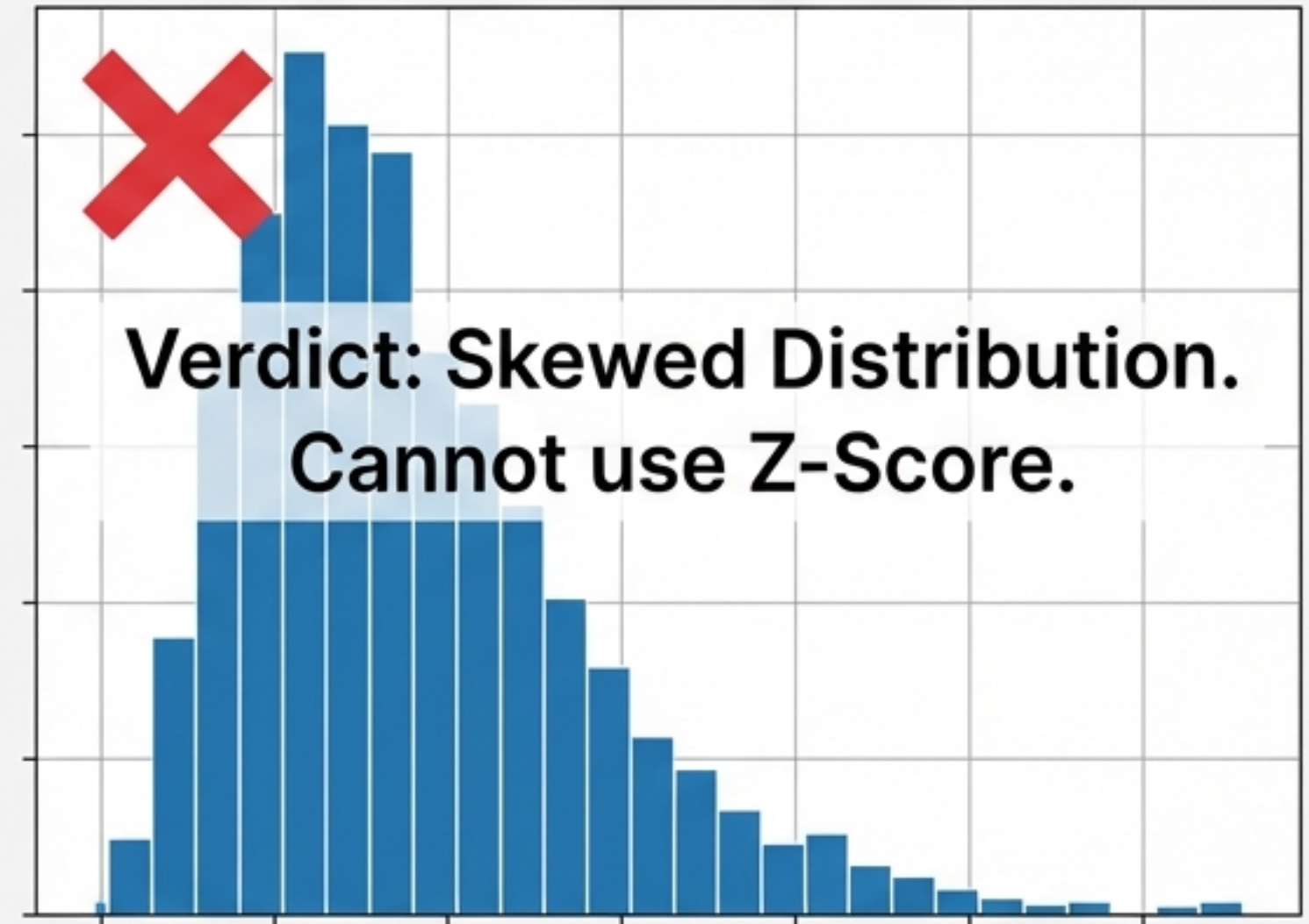
Case Study: Student Placement Data

Visual Inspection of Distributions (N=1000 Students)

Semester Grades (CGPA)



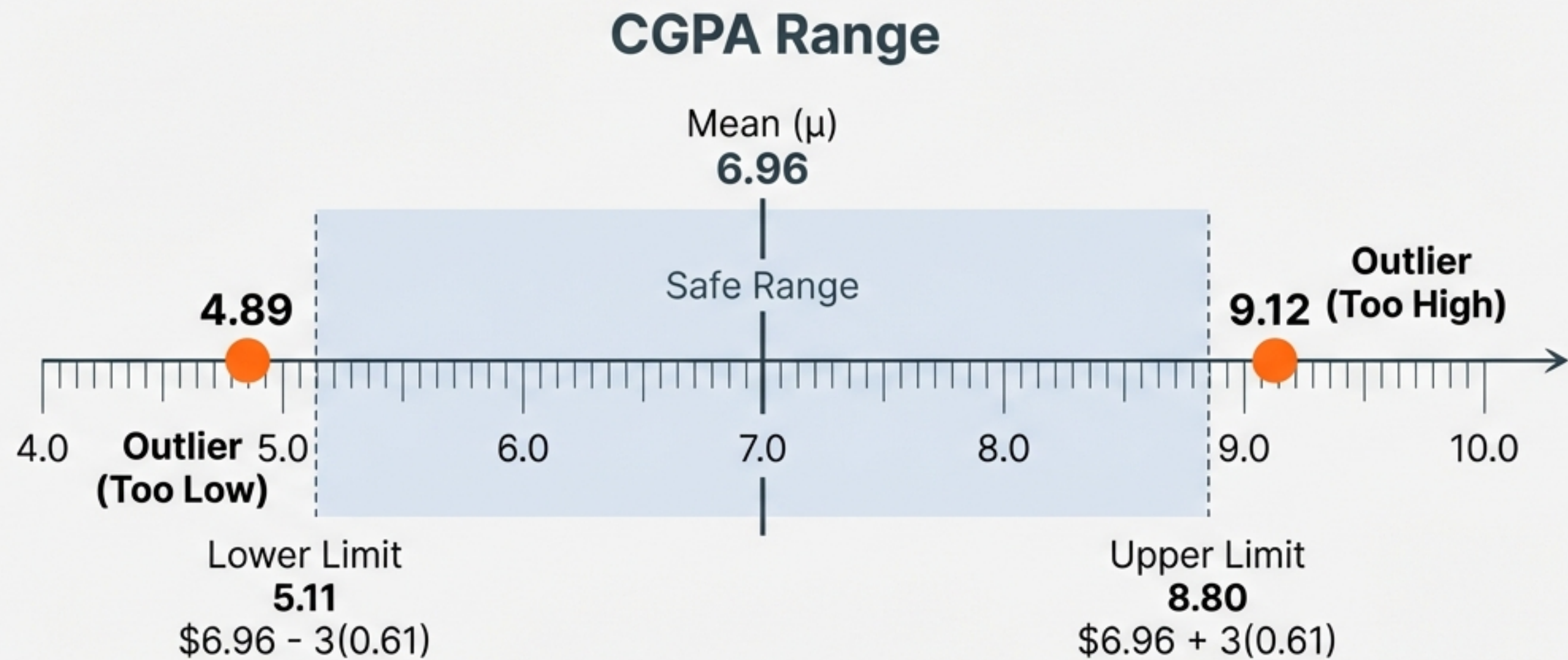
Placement Exam Marks



We will apply outlier detection to the **CGPA** column only.

Detecting the Anomalies

Data Snapshot
Feature: CGPA
Mean (μ): 6.96
Std Dev (σ): 0.61



Total Outliers Detected: 5

Strategy 1: Trimming

Removing rows containing outlier values entirely.



The Math

Initial Count: 1000 Students

Outliers Removed: -5

Final Count: **995 Students**

Critique Box

Pros: Simple implementation.

Cons: Data loss. Risky if dataset has many outliers.

Strategy 2: Capping (Winsorization)

Replacing extreme values with the calculated boundary limits.



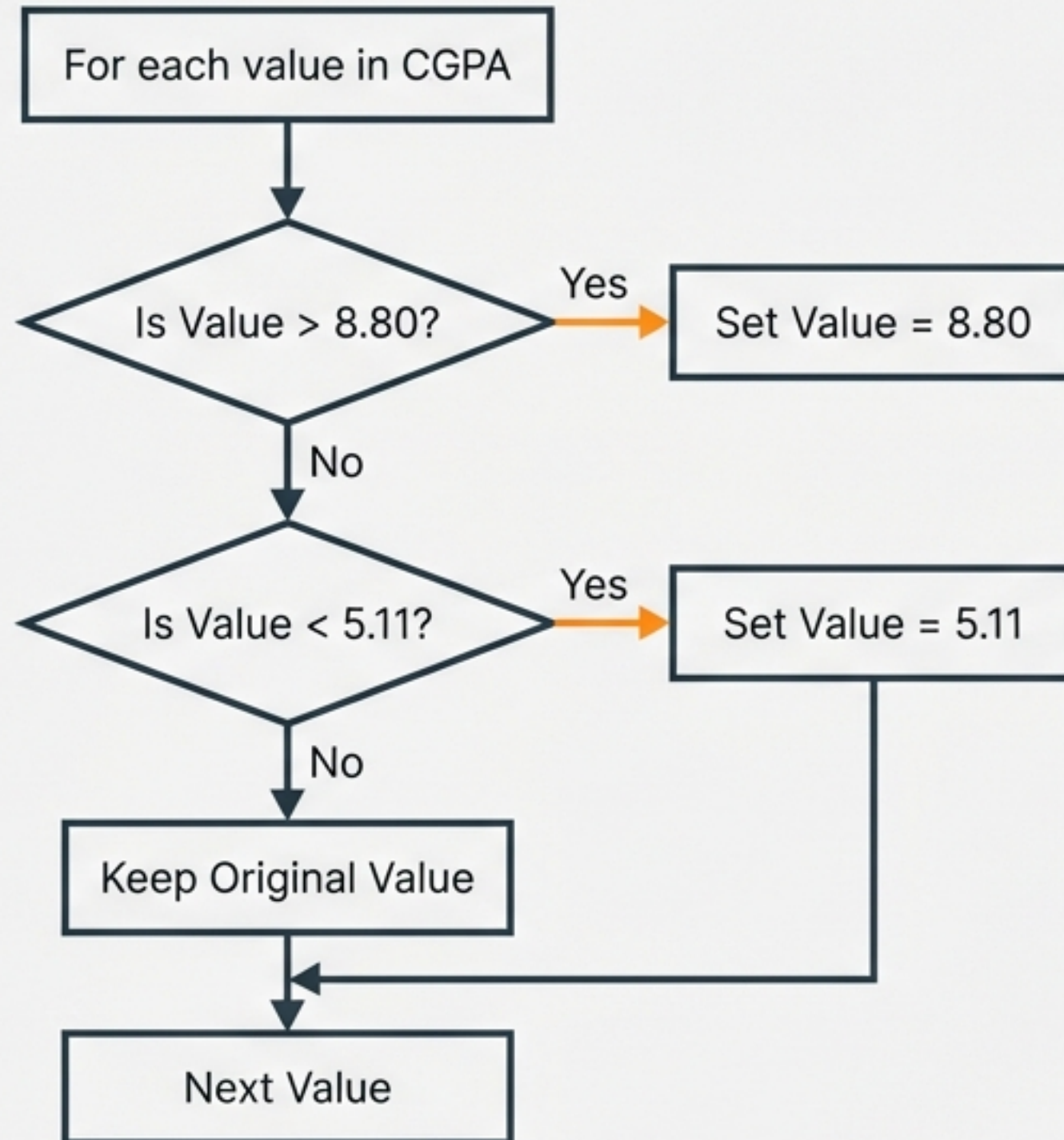
The Result

Initial Count: 1000 Students

Final Count: **1000 Students**

Benefit: Preserves dataset size and pattern.

Implementation Logic: Capping



```
df['cgpa'] = np.where(df['cgpa'] > upper_limit, upper_limit, \
    np.where(df['cgpa'] < lower_limit, lower_limit, df['cgpa']))
```

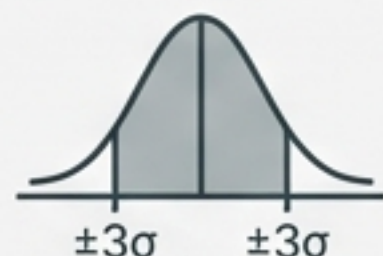

The Results: Before vs. After Capping

Metric	Before Capping	After Capping	Status
Min Value	4.89 ↘	5.11 ✓	Controlled
Max Value	9.12 ↗	8.80 ✓	Controlled
Mean (μ)	~6.96	~6.96	Stable
Std Dev (σ)	~0.61	~0.61	Stable

The Min/Max values are now contained exactly within the boundaries (5.11 - 8.80), yet the statistical properties (Mean/Std Dev) remain intact.

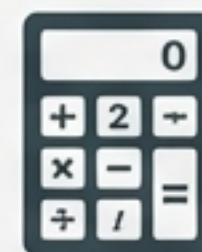
Summary & Key Takeaways

The Rule



Must be **Normal Distribution**.
 $\mu \pm 3\sigma$ covers 99.7%

The Calculation



Upper = $\mu + 3\sigma$
Lower = $\mu - 3\sigma$

The Action



Trim



Cap

Trim for speed/large data.
Cap to preserve data size

The Exception



If Data is Skewed:
Do NOT use Z-Score.
Use IQR / Box Plot method instead.